Among the Clouds

Pupil sheet

Name: _____ **Date:** _____

Safety: Both hands on the box, a metre of clear space, cable slack. Never throw it.

1. What I see on the screen

The box is the **control stick**. It has no windows and no compass — and yet it knows how you are holding it.

The rules: three lives, rings give points, clouds take a life; every five rings it gets faster. The screen shows **Roll** and **Pitch**. After a collision you have a second and a half in which you cannot be hit again — which is why you sometimes pass through a cloud and lose nothing.

The box is held in place and only tilted. If you move it through the air the aeroplane lurches — and that is one of the things to notice today.

2. What I do on my own

First I predict. I do not change my prediction after I play.
I think the aeroplane is controlled by: ☐ pressing something ☐ moving the box through the air ☐ tilting the box
If I turn the box upside down, I think the aeroplane will: _____

because _____

_____

What controls what. I hold the box *in place* and tilt one direction at a time. I draw an arrow for what the aeroplane does.

What I do with the box	What the aeroplane does	Does it stop when I level the box?
Tilt it left		yes / no
Tilt it right		yes / no
Nose up		yes / no
Nose down		yes / no
Turn it upside down		yes / no

The rounds. The score is not the objective — what counts is what I changed between rounds.

Round	Rings	What I changed before the next one
1		
2		
3		

The hard question. When I tilt further, does the aeroplane go *further* or move *faster*? ☐ further ☐ faster

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

The box knows how I hold it because _____

_____

Exit ticket

How does the box know the way you are holding it?

_____